

Gavin McKay – Software Engineer (AI/ML Focus)

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EDUCATION

North Carolina State University

B.S. in Computer Science | Philosophy Minor

Raleigh, NC

Expected Graduation May 2027

Relevant Coursework: Software Fundamentals & Lab, Discrete Math, C & Software Tools, Data Structures & Algorithms, Automata & Languages, Software Engineering, Operating Systems, Intro to AI, Automated Learning & Data Analysis

EXPERIENCE & LEADERSHIP

Vantaca | AI Engineer Intern | Wilmington, NC

May-July 2026

- Contributing to **VantacaCore** – a production SaaS platform powering HOA management for thousands of communities nationwide.
- Multi-Owner Migration:** Contributing to a full-stack migration platform (**ASP.NET Core**, **Azure Functions**, **React/Vite**, **PostgreSQL**) that converts legacy HOA single-owner records into a multi-owner model using AI-assisted name resolution and a transactional push pipeline.
- Platform Defect Resolution:** Triaged and resolved production bugs in the VantacaCore legacy **.NET framework** platform – addressing **SonarQube** code quality gates through full **Azure DevOps** PR review cycles.

Neurotech at NCSU | Lead Software Engineer | Raleigh, NC

Sep 2025-Present

- Engineering Leadership:** Spearheading the software development roadmap for brain-computer interface (BCI) projects, managing a team of developers to integrate electroencephalogram (EEG) data with real-time applications.
- Machine Learning & Signal Processing:** Designing and implementing machine learning algorithms to filter raw EEG signal noise and classify specific brain-wave patterns to trigger digital and physical actions.
- Assistive Technology Development:** Leading the creation of a grid-based, switch-accessible Augmentative and Alternative Communication (AAC) application utilizing eye-tracking technology for children with severe motor impairments.
- Hardware Integration:** Coordinating with neurotech industry partners to integrate high-fidelity EEG headsets for future-phase projects, including brain-controlled robotics (RC cars) and neuro-feedback gaming environments.

PROJECTS

CogniShift– br41n.io BCI Hackathon (3rd Place) | Raleigh, NC

April 2026

- Secured 3rd place and the highest concept grade at the br41n.io BCI hackathon by architecting a system that maps real-time EEG wave data to musical stimuli utilizing Neuropawn headsets and their **BrainFlow repository** for neural signal interfacing.
- Collaborated on an end-to-end signal processing pipeline to isolate specific neural frequencies and translate them into interactive audio environments.

Down Under – cofoundr Hackathon (Most Sustainable Product) | Raleigh, NC

March 2026

- Architected a health-tech mobile platform (**MVP**) designed to alert college-aged women to dangerous drug contraindications between hormonal birth control and common medications.
- Integrated **NLM RxNav** and **openFDA** drug-label **APIs** to power medication search, RxCUI-based interaction checks, and supplemental label-derived interaction text in a Node/Express service.

CageIntel – AI-powered UFC analytics application

2025-Present

- Built a **Python scraper** using **BeautifulSoup** & **Requests** to create a comprehensive dataset of UFC fight stats, including detailed round-by-round data from **ufcstats.com**.
- Trained a **PyTorch** model on the collected data to forecast winners.
- Containerizing the full-stack application to ensure consistent, isolated environments and enable CI/CD.

Password Manager – Python-based password storage application

2022-2023

- Created a Python-based password database application using **MySQL** and **Python** for the full tech stack.
- Engineered the application to store passwords locally via a hidden folder, encrypted using the SHA-256 algorithm, and accessed via a master password.
- Implemented a secure password generator for enhanced security.

SKILLS

Tools: Git/GitHub, Docker, Ollama, Azure, Oracle DB, Kubernetes, MATLAB, MySQL, TensorFlow, PyTorch, OpenCV, Tkinter, pygame, Flask, Tailwind CSS, Selenium, BeautifulSoup, Linux (Debian, Arch, Fedora/Red Hat), Vite, PostgreSQL

Languages: Python, Java, Javascript, Typescript, HTML5, CSS3, C, C#, C++, Bash, Assembly, SQL

Frameworks: React Native, Node.js, Yarn, Django, FastAPI, RestAPI, .NET, Angular

Technical Skills: Software Development Lifecycle, OOP, Machine Learning Algorithms, Data Modeling/Visualization, Software Architecture, Testing & QA, Database Design, DevOps, Problem-Solving, Communication, CI/CD, Server Operations, Linux System Administration, Command Line Operations, Agile development, Plan-driven development